CLICK HERE CLICK HERE

APUL

NEET FLIX

AIM FREE EDUCATION TO ALL

NEET CHANNEL

JOIN OUR TELEGRAM CHANNELS HERE

TEST PAPERS <<< CLICK

TOPPER'S NOTES <<< CLICK

BEST FROM KOTA LECTURES <<< CLICK

QUIZES <<< CLICK

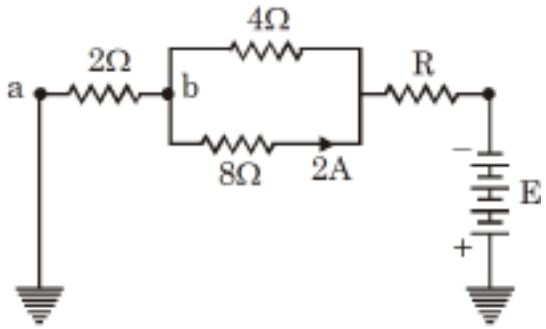
REVISION FLASH CARDS &

MANY MORE <<< CLICK

PHYSICS

1)

In the circuit shown, the current in 2 Ω resistor is



(1) 2 A

(2) 4 A

(3) 6 A

(4) 8 A

Common Content for Question No. 2 to 3

For some isotropic materials, the table of their coefficient of thermal expansions are given. answer the following questions.

Metal	Coefficient of Linear Expansion $\left(\frac{1}{L}\frac{\Delta L}{\Delta T}\right)$	Thermal coefficient of resistivity $\left(\frac{1}{\rho}\frac{\Delta \rho}{\Delta T}\right)$
A	0.0000	0.0002
B	0.0002	0.0002
C	0.0003	0.0002
D	0.0004	0.0000

2) If all metal have same resistance at any temperature then by increasing their temperature by same small amount, their new resistance will be as :-

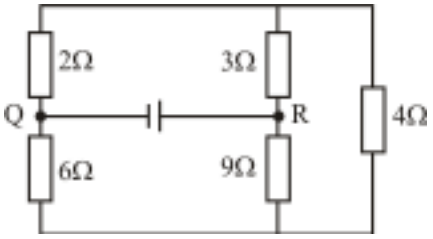
(1) $R_A > R_B > R_C > R_D$

(2) $R_D > R_C > R_B > R_A$

(3) $R_A = R_D > R_C > R_B$

(4) $R_A = R_D = R_B < R_C$

3) Find equivalent resistance (in ohms) across terminals of the battery :-



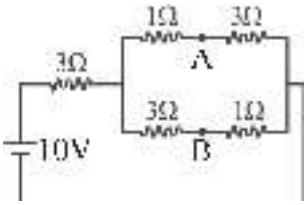
(1) $\frac{20}{3}$

(2) $\frac{15}{4}$

(3) $\frac{16}{3}$

(4) $\frac{21}{4}$

4) In the circuit shown, the potential difference $V_A - V_B$ is :



(1) –2 V

(2) 2 V

(3) 5 V

(4) $\frac{20}{11}$V

5) In the process of measuring current through galvanometer one wish to have 1% of circuit current in the coil. If coil resistance is 99 Ω find shunt resistance used in parallel :-

(1) 0.1 Ω

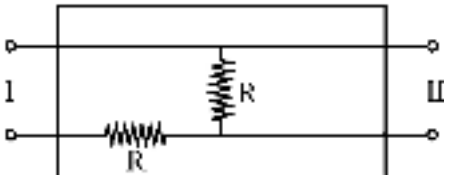
(2) 1.1 Ω

(3) 1.0 Ω

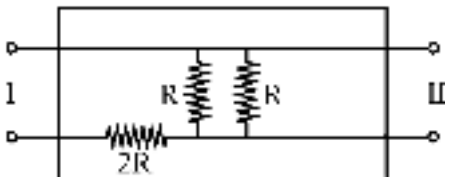
(4) 11.0 Ω

6) If a battery of voltage V is connected across terminals I of the black box as shown in figure, an ideal voltmeter connected to terminals II gives a reading of V/2. If the same battery is connected to terminals II, an ideal voltmeter across terminals I reads V. Assuming that the black box consists only resistances, which of the following can be the possible combinations ?

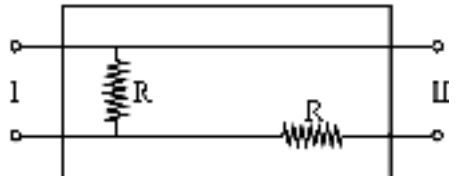
(a)



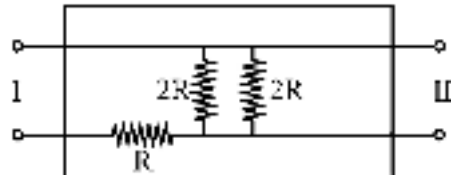
(b)



(c)



(d)



(1) a, c

(2) b, d

(3) b, c

(4) a, d

7) An ammeter and a voltmeter are joined in series to a cell. Their readings are A and V respectively. If a resistance is now joined in parallel with the voltmeter,

(1) both A and V will increase

(2) both A and V will decrease

(3) A will decrease, V will increase

(4) A will increase, V will decrease

8) Two electric bulbs have tungsten filament of same length. If one of them gives 60W and the other 100W, then

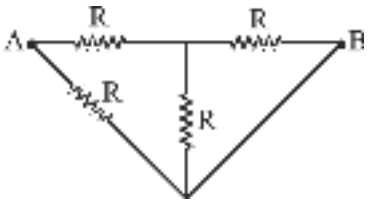
(1) 100W bulb has thicker filament.

(2) 60W bulb has thicker filament.

(3) Both filaments are of same thickness.

(4) It is not possible to get different wattages unless the length are different.

9) Find equivalent resistance between A and B :-



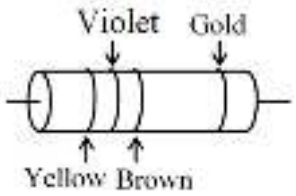
(1) $\frac{3R}{5}$

(2) $\frac{2R}{5}$

(3) $\frac{R}{5}$

(4) $\frac{5R}{3}$

10) A carbon resistor has coloured strips as shown in figure. What its resistance ?



(1) $410\,\Omega \pm 2\%$

(2) $470\,\Omega \pm 5\%$

(3) $420\,\Omega \pm 3\%$

(4) $405\,\Omega \pm 2\%$

11) **Statement-1 :** Conductivity of a metallic conductor decreases with increase in temperature.
Statement-2 : On increasing temperature, collision of electrons becomes more frequent and number of free electrons in the metallic conductor decreases.

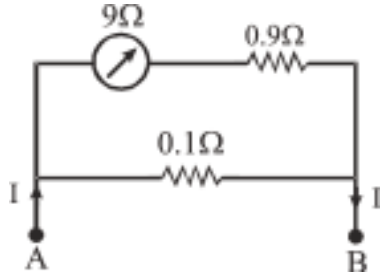
(1) Statement–1 is True, Statement–2 is True ; Statement–2 is a correct explanation for Statement–1

(2) Statement–1 is True, Statement–2 is True ; Statement–2 is not a correct explanation for Statement–1

(3) Statement–1 is True, Statement–2 is False

(4) Statement–1 is False, Statement–2 is True

12) A milliammeter of range 10 mA and resistance 9Ω is joined in a circuit as shown. The metre gives full-scale deflection for current I when A and B are used as its terminals, i.e., current enters at A and leaves at B. The value of I is :-



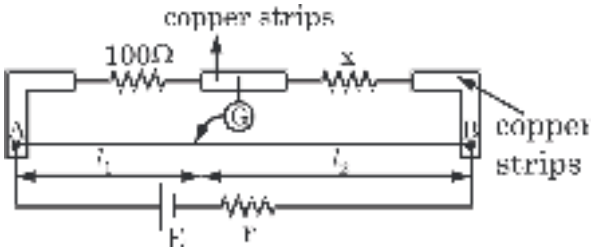
(1) 100 mA

(2) 900 mA

(3) 1 A

(4) 1.1 A

13) In a practical wheat stone bridge circuit as shown, when one more resistance of 100Ω is connected in parallel with unknown resistance 'x', then ratio $\frac{\ell_1}{\ell_2}$ becomes ' 2 '. ℓ_1 is balance length. AB is a uniform wire. Then value of ' x ' must be :



(1) 50 Ω

(2) 100 Ω

(3) 200 Ω

(4) 400 Ω

14) A resistor has colour code of sequence red, yellow, violet, silver. What is value of its resistance :-

(1) $24 \times 10^7 \pm 10\%$

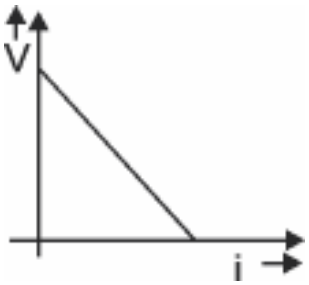
(2) $24 \times 10^7 \pm 5\%$

(3) $42 \times 10^7 \pm 10\%$

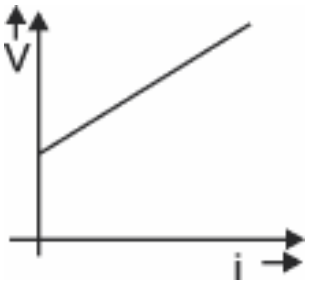
(4) $42 \times 10^7 \pm 5\%$

15) If internal resistance of a cell is proportional to current drawn from the cell. Then the best representation of terminal potential difference of a cell with current drawn from cell will be:

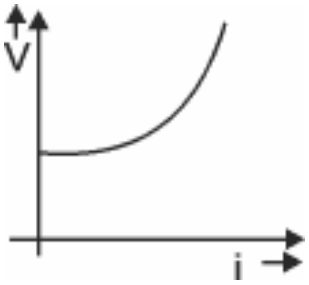
(1)



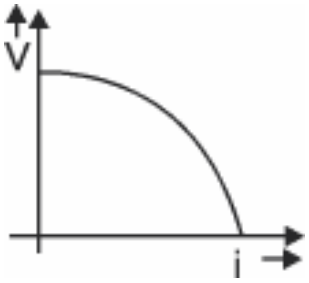
(2)



(3)



(4)



16) The variation of current (I) and voltage (V) is as shown in figure A. The variation of power P with

current I is best shown by which of the following graph

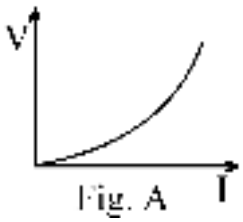
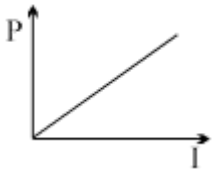
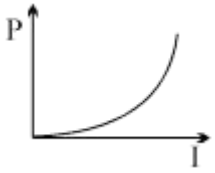
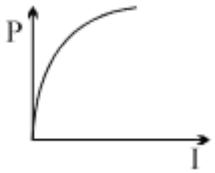
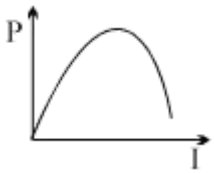


Fig. A

(1)


(2)


(3)


(4)


17) A rectangular parallelopiped is measured as 1 cm × 1 cm × 100 cm. If its specific resistance is 3 × 10^{-7} Ωm, then the resistance between its two opposite rectangular faces will be_____× 10^{-7}Ω.

(1) 5
(2) 9
(3) 3
(4) 7

18) Use the small angle approximation to find approximate value for $\left(\frac{\sin 1^\circ}{\cos 2^\circ}\right)$:-

(1) 1
(2) $\frac{\pi}{30}$
(3) $\frac{\pi}{60}$
(4) $\frac{\pi}{180}$

19) If $(x + y)^2 = 4$, then $\frac{dy}{dx}$ is :

(1) $\frac{-1}{2(x+y)}$

(2) $\frac{1}{2(x+y)}$

(3) 1

(4) –1

20) $y = x^2 - 15x + 56$
for the given equation find correct option.

(a) Minimum value of y is $-\frac{1}{4}$

(b) Minimum value of y is $\frac{1}{4}$

(c) Slope at x = 7.5 is zero

(d) y = 0 for x = 7 and 8

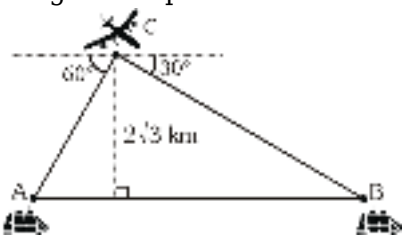
(1) a, b

(2) b, c

(3) a, c, d

(4) Only a

21) The figure shows two ships at points A and B, which are in the same vertical plane with an aeroplane at point C. When the height of the aeroplane is $2\sqrt{3}$ km, the angle of depression to A is 60° and to B is 30°, find the distance between two ships in kilometers.



(1) 4

(2) 8

(3) 6

(4) 2

22) Value of $\int_0^{\pi/2} \cos 3t \ dt$ is

(1) $\frac{2}{3}$

(2) $-\frac{1}{3}$

(3) $-\frac{2}{3}$

(4) $\frac{1}{3}$

23) If vectors $\vec{A} = (\hat{i} + 2\hat{j} + 3\hat{k})m$ and $\vec{B} = (\hat{i} - \hat{j} - \hat{k})m$ represent two sides of a triangle, then the third side can have length equal to :

(a) 3m (b) 5m
(c) 6m (d) 2m

(1) Only (a)

(2) Only (b)

(3) (a), (c)

(4) (a), (b)

24) Two vectors $\vec{a}$ and $\vec{b}$ have magnitudes of 3 units and 4 units, respectively. Angle between $\vec{a}$ and $\vec{b}$ is θ.

List-I		List-II	
(I)	$\vec{a} \cdot \vec{b} = 0$	(P)	$\theta = 0$
(II)	$\hat{a} \cdot \vec{b} = 4$ units	(Q)	$\theta = \pi$
(III)	$\vec{a} \cdot \vec{b} = -12$ units	(R)	$\theta = \frac{\pi}{4}$
(IV)	$\left\|\vec{a} + \vec{b}\right\| = 5$ units	(S)	$\theta = \frac{\pi}{2}$
		(T)	$\theta = \frac{\pi}{3}$

(1) I → S;II → P;III → Q;IV → R

(2) I → S;II → P;III → Q;IV → S

(3) I → R;II → Q;III → P;IV → S

(4) I → Q;II → S;III → R;IV → P

25) Find the resultant of the following two vectors $\vec{A}$ and $\vec{B}$.
$\vec{A}$: 40 units due east and ; $\vec{B}$: 25 units 37° north of west

(1) 25 units 37° north of west

(2) 25 units 37° north of east

(3) 40 units 53° north of west

(4) 40 units 53° north of east

26) A bird moves from point (1, –2) to (4, 2). If the speed of the bird is 10 m/sec, then the velocity vector of the bird is :-

(1) $5\left(\hat{i} - 2\hat{j}\right)$

(2) $5\left(4\hat{i} + 2\hat{j}\right)$

(3) $0.6\hat{i} + 0.8\hat{j}$

(4) $6\hat{i} + 8\hat{j}$

27) If $\vec{A} = 4\hat{i} + 3\hat{j} - 2\hat{k};\ \vec{B} = 6\hat{i} + 8\hat{j} - 4\hat{k};\ \vec{C} = 3\hat{i} + \hat{j} - \hat{k}$. Find $(\vec{A} \times \vec{B}) \cdot \vec{C}$

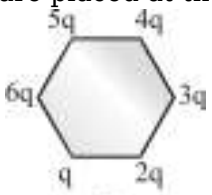
(1) 5

(2) 2

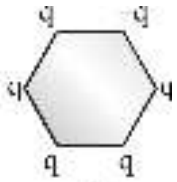
(3) –2

(4) 3

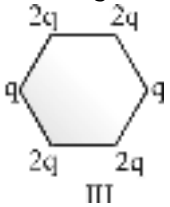
28) Given figures show regular hexagon, the charges are placed at the vertices. In which of the following cases the electric field at the centre is zero.



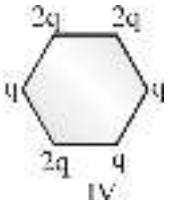
I



II



III



IV

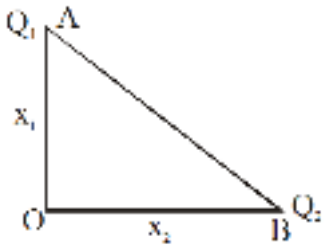
(1) IV

(2) III

(3) I

(4) II

29) Charges Q_1 and Q_2 are at points A and B of a right angle triangle OAB (see figure). The resultant electric field at point O is perpendicular to the hypotenuse, then Q_1/Q_2 is proportional to :



(1) $\frac{x_2^2}{x_1^2}$

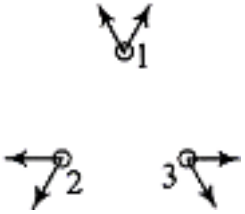
(2) $\frac{x_1^3}{x_2^3}$

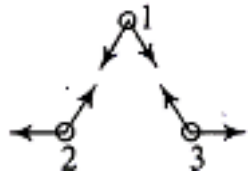
(3) $\frac{x_1}{x_2}$

(4) $\frac{x_2}{x_1}$

30) Which of the following four figures correctly show the forces that three charged particles can exert on each other?

(i) 

(ii) 

(iii) 

(iv) 

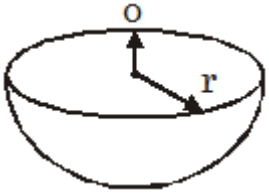
(1) all of the above

(2) none of the above

(3) ii, iii

(4) ii, iii and iv

31) A point charge Q is located just above the centre of the flat face of hemisphere as shown in figure. The electric flux through the flat face and curved face of hemisphere are respectively



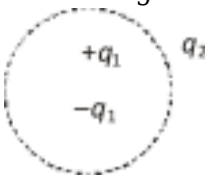
(1) $\frac{Q}{\varepsilon_0}, \frac{-Q}{2\varepsilon_0}$

(2) $\frac{-Q}{2\varepsilon_0}, \frac{Q}{2\varepsilon_0}$

(3) $\frac{Q}{\varepsilon_0}, \frac{-Q}{\varepsilon_0}$

(4) $\frac{-Q}{\varepsilon_0}, \frac{Q}{\varepsilon_0}$

32) Consider the charge configuration and spherical Gaussian surface as shown in the figure. When calculating the flux of the electric field over the spherical surface the electric field will be due to



(1) q_2

(2) Only the positive charges

(3) $+q_1$ and $-q_1$

(4) All the charges

33)

Choose the incorrect statement :
(a) The electric lines of force entering into a Gaussian surface provide negative flux.
(b) A charge 'q' is placed at the centre of a cube. The flux through all the faces will be the same.
(c) In a uniform electric field net flux through a closed Gaussian surface containing no net charge, is zero.
(d) When electric field is parallel to a Gaussian surface, it provides a finite non-zero flux.
Choose the most appropriate answer from the options given below

(1) (c) and (d) only

(2) (b) and (d) only

(3) (d) only

(4) (a) and (c) only

34) A dipole of dipole moment p is kept at the centre of a ring of radius R and charge Q. The dipole moment has direction along the axis of the ring. The resultant force on the ring due to the dipole is :

(1) Zero

(2) $\frac{kPQ}{R^3}$

(3) $\frac{2kPQ}{R^3}$

(4) $\frac{kPQ}{R^3}$ only if the charge is uniformly distributed on the ring

35) The electrostatic potential on the surface of a charged conducting sphere is 100V. Two statements are made in this regard:
S_1 : At any point inside the sphere, electric field intensity is zero.
S_2 : At any point inside the sphere, the electrostatic potential is 100V.
Which of the following is a correct statement ?

(1) S_1 is true but S_2 is false.

(2) Both S_1 & S_2 are false.

(3) S_1 is true, S_2 is also true and S_1 is the cause of S_2.

(4) S_1 is true, S_2 is also true but the statements are independent.

36) An electric dipole moment $\vec{P} = (2.0\hat{i} + 3.0\hat{j})$ μCm is placed in a uniform electric field $\vec{E} = (3.0\hat{i} + 2.0\hat{k}) \times 10^5 NC^{-1}$.
Correct statement are :-
(a) The torque that $\vec{E}$ exerts on P is $(0.6\,\hat{i} - 0.4\,\hat{j} - 0.9\hat{k})$ Nm
(b) The potential energy of the dipole is 0.6 J
(c) The potential energy of the dipole is – 0.6 J
(d) If the dipole is rotated in the electric field, the maximum potential energy of the dipole is 1.3 J

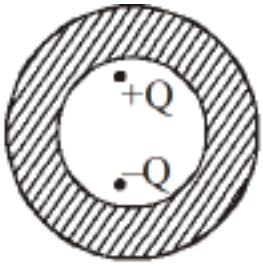
(1) a, b and d

(2) a, c and d

(3) a and b

(4) a and c

37) Shown in the figure are two point charges +Q and – Q inside the cavity of a spherical conductor shell. The charges are kept near the surface of the cavity on opposite sides of the centre of the shell. If σ_1 is the surface charge on the inner surface and Q_1 net charge on it and σ_2 the surface charge on



the outer surface and Q_2 net charge on it then :

(1) $\sigma_1 \neq 0, Q_1 \neq 0$
$\sigma_2 \neq 0, Q_2 \neq 0$

(2) $\sigma_1 = 0, Q_1 = 0$
$\sigma_2 = 0, Q_2 = 0$

(3) $\sigma_1 \neq 0, Q_1 = 0$
$\sigma_2 = 0, Q_2 = 0$

(4) $\sigma_1 \neq 0, Q_1 = 0$
$\sigma_2 \neq 0, Q_2 = 0$

38) In an electric field the potential at a point is given by the following relation $V = \frac{343}{r}$. The electric field at $\vec{r} = 3\hat{i} + 2\hat{j} + 6\hat{k}$ is :

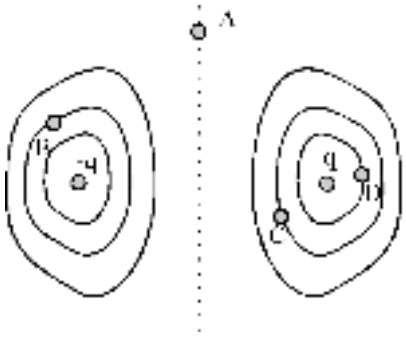
(1) $21\hat{i} + 14\hat{j} + 42\hat{k}$

(2) $3\hat{i} + 2\hat{j} + 6\hat{k}$

(3) $\frac{1}{7}\left(3\hat{i} + 2\hat{j} + 6\hat{k}\right)$

(4) $-\left(3\hat{i} + 2\hat{j} + 6\hat{k}\right)$

39) Figure shows equi-potential surfaces for a two charges system. At which of the labeled points



will an electron have the highest potential energy ?

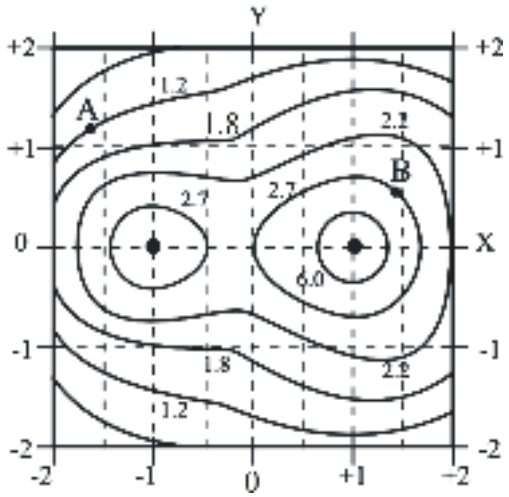
(1) Point A

(2) Point B

(3) Point C

(4) Point D

40) The figure shows equipotential contours for two charges located along the X-axis. Two points A and B have been marked on 1.2 volt and 2.7 volt contours respectively. Choose the correct



statement(s). (a) The charges are in the ratio 1:3.
(b) Both the charges are positive.
(c) Potential energy of a negative charge increases when moved from point A to point B
(d) On X-axis, the field is zero some where between x = –1 and x = +1.

(1) a, b and d

(2) b, c and d

(3) b and c

(4) All statements

41) The potential difference between a certain thundercloud and the ground is 4 MV. During a lightning stroke, 80 C of charge is transferred between the cloud and the ground. The energy dissipated during the stroke is :-

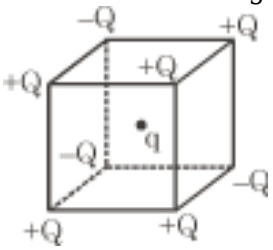
(1) 5×10^{-6} J

(2) 3.2×10^{7} J

(3) 2×10^{5} J

(4) 3.2×10^{8} J

42) Nine point charges are placed on a cube as shown in the figure. The charge q is placed at the body centre whereas all other charges are at the vertices. The electrostatic potential energy of the



system will be :-

(1) Positive

(2) Negative

(3) Zero

(4) Depends on sign and value of q

43) **Assertion (A) :** Under electrostatic condition, the electric field at surface of conductor can have no component parallel to surface.
Reason (R) : The surface of conductor is equipotential and no work is done by electric field to move

the charge on surface.

(1) (A) is True, (R) is True ; (R) is a correct explanation for (A).

(2) (A) is True, (R) is True ; (R) is not a correct explanation for (A).

(3) (A) is True, (R) is False.

(4) (A) is False, (R) is True.

44) Quantization of charge implies

(1) charge exists on particles

(2) there is a minimum permissible magnitude of charge

(3) charge, which is a fraction of coulomb is not possible

(4) none of the above

45) Potential energy of a system comprising of point charges is U_1.When a charge q is added in the system without disturbing other charges, the potential energy becomes U_2.The potential of the point where the charge q is placed in the system is

(1) $\frac{U_2 - U_1}{q}$

(2) $\frac{U_1 - U_2}{q}$

(3) $\frac{U_1 + U_2}{2q}$

(4) $\frac{U_2 - U_1}{2q}$

CHEMISTRY

1) $K_6\left[(CN)_5Co-O-O-Co(CN)_5\right] \xrightarrow{\text{Oxidises}}$ (x)

$K_5\left[(CN)_5Co-O-O-Co(CN)_5\right]$ (y)

In both complexes Co have $t_2g^6\,eg^0$ configuration the bond energy (B.E.) in x and y is-

(1) B.E. of (O–O) in y < B.E. of (O–O) in x

(2) B.E. of (O–O) in x < B.E. of (O–O) in y

(3) B.E. of (O–O) in x = B.E. of (O–O) in y

(4) B.E. of (O–O) in x and B.E. of (O–O) in y can't be predicted.

2) Consider the following statements and arrange in the order of true/false as given in the codes. S_1 : Both $[Co(ox)_3]^{3-}$ and $[CoF_6]^{3-}$ are paramagnetic.
S_2 : $CoCl_3$ $3NH_3$ complex is non-conducting
S_3 : The number of possible isomers for the complex $[Pt(NO_2)\,(py)\,(OH)\,(NH_3)]$ is six.
S_4 : The oxidation state of iron in brown ring complex $[Fe(H_2O)_5NO]\,SO_4$ is + II where NO is NO^+

(1) F T T T

(2) T T T T

(3) F T T F

(4) T T T F

3) A, B and C are possible isomers of $CoCl_3.6H_2O$. A shows maximum electrical conductance and C has maximum percentage of non ionisable Cl^-. Then incorrect statement is :-

(1) No weight loss, when A is treated with conc. H_2SO_4

(2) Order of number of moles of AgCl ppt on adding excess $AgNO_3$ solution is A > B > C

(3) Complex B is $[Co(H_2O)_5Cl]Cl_2.H_2O$

(4) Order conductivity of complex containing chloride ion A < B < C

4) For the complex, $[Fe^{II}(en)_3]_n\ [Co^{III}(NH_3)_2(NO_2)_4]_2$ the value of n is :-

(1) 1

(2) 2

(3) 3

(4) 4

5) **Assertion (A)** : In metal carbonyls, the metal to ligand bonding creates a synergic effect which strengthens the bond between CO and the metal.
Reason (R) : The metal carbon bond in metal carbonyls possess both σ and π characters.
Choose the correct answer from the options given below :

(1) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(2) (A) is true but (R) is false.

(3) (A) is false but (R) is true.

(4) Both (A) and (R) are true and R is correct explanation of (A).

6) Correct Statement is _____.
(A) Complex $[Pt(NH_3)_2Cl_2]$ show geometrical isomerism
(B) Optical isomers are mirror images that can not be superimposed on one another. These are called enantiomers.
(C) Chlorophyll is a complex compound of magnesium
(D) Ethylenediamine is a monodentate ligand.

(1) (A),(B),(C),(D) are correct options

(2) (A),(B),(C) are correct options

(3) (A),(B),(D) are correct options

(4) (A),(D) are correct options

7) Which is correct match :-

Complex	**No. of Chilate rings**
(1) $[Co(ox)_2Cl_2]^{-3}$	4
(2) $[Ni(dmg)_2]$	4
(3) $[Ca(EDTA)]^{-2}$	6

(4) Brown ring complex 1

(1) 1

(2) 2

(3) 3

(4) 4

8) (Valence Bond Theory) The coordination complex $[Cu(OH_2)_6]^{2+}$ has one unpaired electron. Which of the following statements are true ?
(1) The complex is octahedral
(2) The complex is an outer orbital complex
(3) The complex is d^2sp^3 hybridized
(4) The complex is diamagnetic
(5) The coordination number is 6

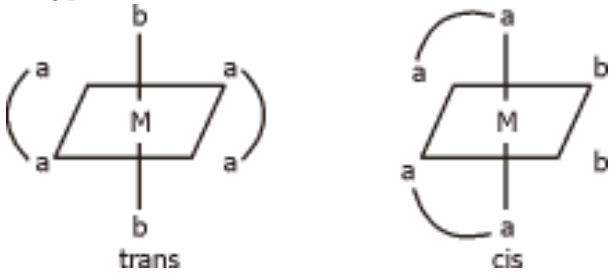
(1) 1, 4

(2) 1, 2, 5

(3) 2, 3, 5

(4) 2, 3

9) Which of the following statements is correct regarding the chirality (optical isomerism) of the cis and trans isomer of the type $M(aa)_2b_2$ (M stands for a metal, (aa) is bidentate ligand and b is a monodentate ligand)



(1) The trans form is achiral and optically inactive while the cis form is chiral and exists in two enantiomeric forms

(2) Both cis and trans forms are achiral and optically inactvie

(3) The trans form is chiral and exists in two enantiomeric forms while the cis form is achiral and optically inactive

(4) Both the cis and trans form are chiral and each exist in two enantiomeric form

10) The IUPAC name for $K_2[Cr(CN)_2O_2(O_2)NH_3]$ is :-

(1) Potassium amminedicyanotetraoxo chromium (III)

(2) Potassium amminedicyanodioxygendioxo chromate (IV)

(3) Potassium amminedicyanosuperoxoperoxo chromate (III)

(4) Potassium amminedicyanodioxoperoxo chromate (V)

11) Which is wrong statement -

(1) $[Co(NH_3)_6]$ $[Cr(CN)_6]$, $[Pt(NH_3)_4]$ $[CuCl_4]$ shows co-ordination isomerism

(2) Correct relation will be - $\Delta_0 > \Delta_{sp} > \Delta_t$

(3) In octahedral low spin complex having 5 electrons in d subshell the number of unpaired electrons is 1

(4) $[PtCl_4]^{-2}$ have dsp^2 hybridisation

12) Which of the following are inner orbital complex (i.e. involving d^2sp^3 hybridisation) and is paramagnetic in nature?

(1) $[Mn(CN)_6]^{3-}$, $[Fe(CN)_6]^{3-}$, $[Co(C_2O_4)_3]^{-3}$

(2) $[MnCl_6]^{3-}$, $[FeF_6]^{3-}$, $[CoF_6]^{3-}$

(3) $[Mn(CN)_6]^{3-}$, $[Fe(CN)_6]^{3-}$

(4) $[MnCl_6]^{3-}$, $[Fe(CN)_6]^{3-}$, $[Co(C_2O_4)_3]^{-3}$

13) Which is incorrect :

(1) Conductivity = $[Co(NH_3)_3Cl_3] < [Cr(NH_3)_5Cl]Cl < [Co(NH_3)_5Cl]Cl_2 < [Co(NH_3)_6]Cl_3$

(2) cis form of $[M(AA)_2X_2]^{+n}$ =optically inactive

(3) $[CoF_6]^{-3} = t_{2g}^4 e_g^2$ $[Fe(CN)_6]^{-4} = t_{2g}^6 e_g^0$ $[Cu(NH_3)_6]^{+2} = t_{2g}^6 e_g^3$

(4) Δ_0 order = $[Cr(Cl)_6]^{-3} < [Cr(NH_3)_6]^{+3} < [Cr(CN)_6]^{-3}$

14) Match the column

	Column-I Compound		**Column-II Property**
(a)	$Mn_2(CO)_{10}$	i.	Bridged by two CO group
(b)	$Co_2(CO)_8$	ii.	Two square pyramidal units
(c)	σ & π character	iii.	Vacant antibonding π* orbital
(d)	Metal– carbon π bond	iv.	Metal–carbon bond of metal carbonyls

(1) a - i; b - ii; c - iii; d - iv

(2) a - ii; b - i; c - iii; d - iv

(3) a - ii; b - i; c - iv; d - iii

(4) None

15) Extraction of metals or other process is through the complex formation
(I) Cyanide process
(II) Mond's process
(III) Photographic fixing process
Complexes formed in these methods are -

	I	**II**	**III**
(1)	$[Ag(NH_3)_2]Cl$	$Ni(CO)_4$	$[Ag(CN)_2]^-$

(2)	$[Ag(CN)_2]^-$	$Ni(CO)_4$	$[Ag(S_2O_3)_2]^{3-}$
(3)	$[Ag(CN)_2]^-$	$[Ag(S_2O_3)_2]^{-3}$	$Ni(CO)_4$
(4)	$[Cd(CN)_4]^{2-}$	$Ni(CO)_4$	$[Ag(S_2O_3)_2]^{3-}$

(1) 1

(2) 2

(3) 3

(4) 4

16) Consider following properties of the noble gases.
(1) They readily form compounds which are colourless.
(2) They generally do not from ionic compounds.
(3) Xenon has variable oxidation state in its compounds.
(4) The smaller He and Ne do not from clathrate compounds.
Select correct properties.

(1) 1, 2, 4

(2) 2, 3, 4

(3) 1, 3, 4

(4) All

17) Which element shows allotropy ?

(1) Phosphorous

(2) Arsenic

(3) Nitrogen

(4) 1 & 2 both

18) **Statement-I :** The common oxidation state of group 15 elements it +3 and +5.
Statement-II : Nitrogen react with F_2 to form NF_3 and NF_5.

(1) Both are correct.

(2) Statement-I is correct and Statement-II incorrect.

(3) Statement-I is incorrect and Statement-II correct.

(4) Both are incorrect.

19) Which of the following reaction gives an oxyacid ?

(1) 2 moles of perchloric acid $\xrightarrow{-H_2O}$

(2) 1 mole of sulphurous acid $\xrightarrow{-H_2O}$

(3) 2 moles of sulphuric acid $\xrightarrow{-H_2O}$

(4) 2 moles of nitric acid $\xrightarrow{-H_2O}$

20)

Which of the following compound consist of peroxide bond.

(1) Perchloric acid

(2) Permagnic acid

(3) O_2F_2

(4) Caro's acid (H_2SO_5)

21) Which of the following over lapping in involved in Al – Cl – Al of Al_2Cl_6 ?

(1) $sp^2 - sp^3 - sp^2$

(2) $sp^3 - sp^3 - sp^3$

(3) $sp^3 - sp^3d - sp^3$

(4) $sp^2 - sp^2 - sp^2$

22) $A + H_2O \rightarrow B + HCl$
$B + H_2O \rightarrow C + HCl$
Compound A, B, C are respectively will be :

(1) PCl_5, $POCl_3$, H_3PO_3

(2) PCl_5, $POCl_3$, H_3PO_4

(3) $SbOCl_2$, $POCl_3$, H_3PO_3

(4) PCl_3, $POCl_3$, H_3PO_4

23) Match the following.

Column I (Species)		**Column II (Type of dimerisation)**	
(a)	$(BeH_2)_2$	(p)	3-centre-4e$^-$ bond
(b)	$Al_2(CH_3)_6$	(q)	4-centre-3e$^-$ bond
(c)	$(BeCl_2)_2$	(r)	3-centre-2e$^-$ bond
(d)	$(AlCl_3)_2$	(s)	1-centre-1e$^-$ bond

(1) a - p ; b - q ; c - s ; d - r

(2) a - q ; b - q ; c - r ; d - s

(3) a - r ; b - r ; c - p ; d - p

(4) a - p ; b - p ; c - r ; d - r

24) Consider the following statements
(a) Both rhombic and monoclinic sulphur have S_8 molecules
(b) The S_8 molecule may have crown shape
(c) Monoclinic sulphur are soluble in CS_2
Which is correct :-

(1) Only a and c

(2) Only a, b

(3) Only b

(4) All a, b and c

25) **Statement-I :** SRP of F_2 is much higher than that of Cl_2.
Statement-II : I_2O_5 is oxidising agent and it is used in the estimation of carbon mono oxide.

(1) Both are correct.

(2) Statement-I is correct and Statement-II is incorrect.

(3) Statement-I is incorrect and Statement-II is correct.

(4) Both are incorrect.

26) Which is correct order for group 13 element?

(1) B < Al < Ga < In < Tl (Atomic Radius)

(2) B > Al > Ga > In > Tl (Ionisation energy)

(3) B > Al > Ga > In > Tl (Electronegativity)

(4) B < Al < Ga < In < Tl (Density)

27) $Cl_2 + H_2O \rightarrow x + y$ is ''x'' shows bleaching property. Then select incorrect statement :

(1) Oxidation state of Cl in x and y is +1 and –1 respectively.

(2) x is permanent bleach

(3) x is disproportionate on heating

(4) x bleaches by reducing action.

28) Find correct order :

(1) Density ⇒ α-S = β-S

(2) B.P. ⇒ $O_2 > S_8$

(3) Reducing nature ⇒ $H_2O > H_2Se$

(4) Oxidising property ⇒ $TeO_2 > SO_2$

29) Select the correct order of given prop.

(1) $H_2O > H_2Te > H_2Se > H_2S$: M.P. and B.P.

(2) $H_2Te > H_2Se > H_2S > H_2O$: Acidic strength

(3) $H_2O > H_2S > H_2Se > H_2Te$: Dipole moment

(4) All of these

30) Match the Column -

Column-I (Gas)		**Column-II (Use)**	
(1)	He	(p)	Discharge tube

(2)	Ne	(q)	Gas-cooled nuclear reactor
(3)	Ar	(r)	Used in light bulbs
(4)	Xe	(s)	Are welding of metals or alloys

(1) 1-q, 2-p, 3-s, 4-r

(2) 1-p, 2-q, 3-r, 4-s

(3) 1-r, 2-s, 3-p, 4-q

(4) 1-s, 2-r, 3-q 4-p

31) Which pair is isostructure and possesses same number of lone pair of electron on central atom ?

(1) IF_5^{+2} and $XeOF_4$

(2) NH_3 and ClO_3^-

(3) $SnCl_2$ and ClO_4^-

(4) $AlCl_3$ and SO_2

32) Molecule showing angle greater than observed in tetrahedral structure :-

(1) NH_3

(2) Cl_2O

(3) OF_2

(4) H_2O

33) Correct order of ionic character :-

(1) $CdCl_2 < SnCl_2 < CaCl_2$

(2) $KCl < AgCl$

(3) $ZnCl_2 > CdCl_2 < HgCl_2$

(4) $LiF < Li_2O < Li_3N$

34) In which of the following ionization processes, the bond order has increased and the magnetic behaviour has changed ?

(1) $N_2 \rightarrow N_2^+$

(2) $C_2 \rightarrow C_2^+$

(3) $NO \rightarrow NO^+$

(4) $O_2 \rightarrow O_2^+$

35) Which is not stable ?

(1) KHF_2

(2) KI_3

(3) $CH_3–CH(OH)_2$

(4) $Cl_3C–CH(OH)_2$

36) Which of the following statement is true :-
(A) PH_5, NCl_5 and $BiCl_5$ do not exist
(B) I_3^+ has bent shape
(C) XeF_4 is a non polar molecule
(D) CO and C_2^{-2} do not have same bond order.

(1) A, B, D
(2) A, B, C
(3) B, C, D
(4) B,C

37) Which of the following molecular species is not linear.

(1) $(CN)_2$
(2) OCN^-
(3) XeF_2
(4) S_3^{-2}

38) Which of the following has highest melting point?

(1) NaCl
(2) SiC
(3) NaBr
(4) NaF

39) Which of the order is correct ?

(1) $ClO^- < ClO_2^- < ClO_3^-$ (Bond order)
(2) $CO_3^{-2} > CO > CO_2$ (Bond strength)
(3) $PO_4^{-3} > SO_4^{-2}$ (no. of coordinate bonds)
(4) $C_3O_2 > C(CN)_4$ (no. of π bond)

40) Molecular shape of XeF_3^+ ion and XeF_5^+ ion are respectively :-

(1) Square pyramidal, T-shaped
(2) T-shape, square pyramidal
(3) See-saw, square pyramidal
(4) Square pyramidal, see-saw

41) According to VBT molecule got stability by :-

(1) Completing it's octet
(2) Hybridisation

(3) Pairing it's unpaired e^-

(4) Donating e^-

42) If E = the number of lone pairs on Xe.
B = the number of bonding pairs
S = shape of the molecule,
then what is the correct set of E, B and S of XeF_4 ?

	E	B	S
(1)	3	3	Octahedral
(2)	3	3	square planer
(3)	2	4	square planer
(4)	4	2	square planer

(1) 1

(2) 2

(3) 3

(4) 4

43) Incorrect order of bond dissociation energy will be

(1) H – H > Cl – Cl > Br – Br

(2) Si – Si > C – C > H – H

(3) C – C > N – N > O – O

(4) H – Cl > H – Br > H – I

44) **Assertion :-** Boiling point of H_2SO_4 is more than HNO_3.
Reason :- H_2SO_4 have more extent of intra molecular H-bonding than HNO_3.

(1) Both assertion and reason are true but reason is not correct explanation of assertion.

(2) Assertion is true but reason is false.

(3) Assertion is false but reason is true.

(4) Both assertion and reason are true but reason is a correct explanation of assertion.

45) Match the column.

	Column-I (order)		**Column-II (Properties)**
(A)	$N_2 > N_2^+$	(P)	No. of σ bond
(B)	$B_2 > B_2^+$	(Q)	No. of unpaired e^-
(C)	$O_2^{2-} > O_2$	(R)	Bond order
(D)	$H_2 > H_2^+$	(S)	No. of A.B.M.O e^-s

(1) (A)-P,R (B)-Q,S (C)-S (D)-P,Q,R,S

(2) (A)-Q,R (B)-Q,R (C)-P,Q (D)-Q,R,S

(3) (A)-P,Q,R (B)-P,Q,R (C)-Q,R (D)-R,S

(4) (A)-P,R (B)-Q,R (C)-S (D)-P,R

BIOLOGY

1) Select the correctly matched option from the column I and column II :

	Column I		**Column II**
(a)	*Ramapithecus*	(i)	Lived in East-African grasslands
(b)	*Australopithecus*	(ii)	Probably ate meat
(c)	*Homo habilis*	(iii)	Existed about 15 mya
(d)	*Homo erectus*	(iv)	First human like being

Options :

(1) a-i, b-ii, c-iii, d-iv

(2) a-iii, b-ii, c-i, d-iv

(3) a-i, b-ii, c-iv, d-iii

(4) a-iii, b-i, c-iv, d-ii

2) Match the columns

	Column-I		**Column-II**
(a)	Lamarck	(i)	Mutation theory
(b)	Darwin	(ii)	Use and disuse of organs
(c)	Hugo de vries	(iii)	Origin of life
(d)	Oparin	(iv)	Natural selection

(1) a – ii, b –iv, c – i, d – iii

(2) a – i, b – ii, c – iii, d – iv

(3) a – iii, b –i, c – ii, d – iv

(4) a – ii, b –i, c – ii, d – iii

3) **Assertion :** Lichens can be used as pollution indicators
Reason : Lichens will not grow in areas that are pollution free

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true but Reason is false

(4) Both Assertion and Reason are false

4) **Assertion :** The fitness according to Darwin, refers ultimately and only to reproductive fitness.

Reason : Hence those who are better fit in an environment leave more progeny than others

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

5)

Read two statements A and B carefully and select the correct option regarding them:
(A) The first form of life on earth was chemoheterotrophic and anaerobic.
(B) First photosynthetic organisms on earth did not release oxygen in the environment.
Options:

(1) Statement (A) is correct and statement (B) is incorrect
(2) Statement (B) is correct and statement (A) is incorrect
(3) Both statements are incorrect
(4) Both statements are correct

6) **Statement-I :** Evolution is not a directed process in the sense of determinism.
Statement-II : Evolution is a stochastic process based on chance events in nature and chance mutation in the organisms.

(1) Both Statement-I and Statement-II are incorrect
(2) Statement-I is correct but Statement-II is incorrect
(3) Statement-I incorrect and Statement-II is correct
(4) Both Statement-I and Statement-II are correct

7) The chronological order of human evolution from early to the recent is –

(1) Ramapithecus → Dryopithecus → *Homo erectus* → *Homo habilis*
(2) *Dryopithecus* → *Ramapithecus* → Neanderthal man → *Homo erectus*
(3) *Ramapithecus* → *Australopithecus* → *Homo habilis* → *Homo erectus*
(4) *Homo habilis* → *Ramapithecus* → *Homo erectus* → *Dryopithecus*

8) The correct sequence of periods from most ancient to most recent is :-

(1) Triassic → Jurassic → Cretaceous → Tertiary → Permian
(2) Permian → Jurassic → Triassic → Tertiary → Quaternary
(3) Permian → Triassic → Cretaceous → Jurassic → Tertiary
(4) Permian → Triassic → Jurassic → Cretaceous → Tertiary → Quaternary

9) Choose incorrect statement :

(1) Fossils are remains of hard parts of life forms found in rocks
(2) Rocks form sediments and a cross section of earth's crust indicates the arrangement of sediments one over the other during the long history of earth

(3) A study of fossils in different sedimentary layers indicate the geological period in which they existed

(4) Different aged rock sediments always contain fossils of same life forms

10) Which of the following had the smallest brain capacity ?

(1) *Homo sapiens*

(2) *Homo neanderthalensis*

(3) *Homo habilis*

(4) *Homo erectus*

11) Darwin’s finches are a good example of :

(1) Adaptive convergence

(2) Divergent evolution

(3) Parallel evolution

(4) Convergent evolution

12) Oparin of Russia and Haldane of England proposed that the first form of life could have come from pre-existing ___(A)____ organic molecules and that formation of life was preceded by ___(B)____.

(1) A - Living, B - Chemical evolution

(2) A - Non-living, B - Organic evolution

(3) A - Non-living, B - Chemical evolution

(4) A - Living, B - Inorganic evolution

13)

Which of the following factors do not affect the genetic equilibrium ?

(1) Mutation

(2) Gene flow

(3) Random mating

(4) Genetic drift

14) Which statement is **correct** for industrial melanism?

(1) It is a directional natural selection where median population is selected.

(2) It is a directional natural selection where either of the two extremities are selected at particular time.

(3) It is a stabilising natural selection where median population is selected.

(4) It is a stabilising natural selection where one of the extremities is selected.

15) The theory of spontaneous generation stated that :

(1) life arose from living forms only

(2) Life can arise from both living and non-living

(3) Life can arise from non-living things only

(4) Life arises spontaneously, neither from living nor from the non-living

16) Choose **incorrect** option regarding analogous organs in animals :-

(1) Fore limbs of mammals

(2) Wings of butterfly and birds

(3) Eye of octopus and mammals

(4) Flippers of penguins and dolphins

17) Which human ancestor painted beautiful paintings on cave wall and developed pre historic cave art 18000 years ago ?

(1) *Homo habilis*

(2) Cromagnon man

(3) Java man

(4) *Australopithecus*

18) In a population of 1000 individual, 400 individual are tall, 200 individual are medium, 400 individual are short. If tall and short individual survives, then which type of selection operates in population?

(1) Sexual selection

(2) Stabilising selection

(3) Directional selection

(4) Disruptive selection

19) Match the blood vessels of human heart listed under column-I with the functions given under Column-II; Choose the answer which given the correct combination of the alphabets of the two columns

Column I (Blood Vessel)		**Column II (Function)**	
(A)	Superior vena cava	(p)	Carries deoxygenated blood to lungs
(B)	Inferior vena cava	(q)	Carries oxygenated blood to lungs
(C)	Pulmonary artery	(r)	Bring deoxygenated blood from lower parts of the body to the right atrium
(D)	Pulmonary vein	(s)	Brings oxygenated blood to the left atrium

		(t)	Brings deoxygenated blood from upper parts of the body into the right atrium

(1) A = t, B = p, C = r, D = q

(2) A = t, B = r, C = p, D = s

(3) A = s, B = t, C = r, D = p

(4) A = t, B = s, C = r, D = p

20)

Rh –ve person donated blood to Rh +ve person for the second time then. :-

(1) Rh –ve person will die

(2) Rh +ve person will die

(3) Rh +ve blood starts reacting to Rh –ve blood

(4) Nothing happens to Rh +ve person.

21) Read the following properties of blood corpuscles:-
(a) They are colourless due to lack of haemoglobin
(b) They are non-nucleated
(c) They are relatively lesser in number which averages 6000-8000 mm^{-3} of blood
(d) They have long life span approx 120 days.
Out of these which properties are not shown by leukocytes ?
Options :-

(1) 'a' and 'b'

(2) 'b' and 'c'

(3) 'b' and 'd'

(4) 'c' and 'd'

22) **Assertion :** Reduction in the number of thrombocytes can lead to clotting disorder.
Reason : Platelets can release a variety of substances which are involved in coagulation of blood.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

23)

Match List I with List II.

	List I		**List II**
(A)	P - wave	(I)	Beginning of systole
(B)	Q - wave	(II)	Repolarisation of ventricles

(C)	QRS complex	(III)	Depolarisation of atria
(D)	T - wave	(IV)	Depolarisation of ventricles

Choose the correct answer from the options given below :

(1) A-IV, B-III, C-II, D-I

(2) A-II, B-IV, C-I, D-III

(3) A-I, B-II, C-III, D-IV

(4) A-III, B-I, C-IV, D-II

24) Platelets are formed from :-

(1) Megakaryoblast

(2) Macrophage

(3) Megakaryocyte

(4) RBC

25) Read following statements carefully and choose how many are correct :-
A. Neutrophil and monocyte are phagocytic.
B. Plasma constitute nearly 55% of blood.
C. Basophils secretes histamin, serotonin and heparin
D. Blood contains 1.5-3.5 lakhs platelets mm^{-3}

(1) A, C, D only

(2) A, B, D only

(3) B, C, D only

(4) All A, B, C, D

26) Identify the correctly matched structure :-

(1) Neutrophils → Bilobed nucleus

(2) Eosinophils → Nucleus absent

(3) Monocytes → Kidney shaped nucleus

(4) Erythrocytes → Horse shoe shaped nucleus

27) Which statement is incorrect regarding first heart sound ?

(1) It is produced at the beginning of ventricular systole

(2) It is associated with the closure of the tricuspid and bicuspid valves

(3) It is called "LUBB"

(4) It is high pitch sound for short period

28) Pain in heart muscles is :-

(1) Cardiac arrest

(2) Coronary artery disease

(3) Angina pectoris
(4) Myocarditis

29) SAN is present in :-

(1) Right upper corner of the left atrium
(2) Left upper corner of the right atrium
(3) Right upper corner of the right atrium
(4) Left upper corner of the left atrium

30) Which of the following animals complete double circulation is found?
Human, Fishes, Crocodile, Calotes, Rana, Birds, Pteropus, Aptenodytes

(1) 4
(2) 5
(3) 7
(4) 3

31) About 1000 ml of air is always remain inside the human lungs. It is described as :-

(1) Tidal volume
(2) Residual volume
(3) Expiratory reserve volume
(4) Inspiratory reserve volume

32) Select the correct combination :-

(1) IC = TV + ERV
(2) FRC = ERV + IRV + TV
(3) RV = IRV
(4) VC = ERV + IRV + TV

33) In the given table choose the correct value of partial pressure :-

Respiratory gas	Alveoli	Deoxygenated blood	Oxygenated blood	Tissue
O_2	104	b	95	d
CO_2	a	45	c	45

	a	b	c	d
(1)	40	40	40	40
(2)	45	45	95	45
(3)	45	40	45	95

(4)	40	95	40	45

(1) 1
(2) 2
(3) 3
(4) 4

34) What is the function of pleural fluid?

(1) reduces friction on the lung surface
(2) protection from external shocks
(3) provide moisture
(4) all of the above

35) How many cellular layers are present in the diffusion membrane in lungs ?

(1) one
(2) two
(3) three
(4) four

36) Which of the following statement is incorrect about lung alveoli ?

(1) lined by simple squamous epithelium
(2) primary sites of gaseous exchange
(3) partial pressure of CO_2 in alveoli is 46 mmHg
(4) partial pressure of O_2 in alveoli is 104 mmHg

37) In rabbit alveolar ducts originate from :-

(1) Bronchi
(2) Trachea
(3) Bronchiole
(4) Respiratory bronchiole

38) How many organism in the list given below do respiration by general body surface.
Insects, Sponges, Flatworm, Fishes, Birds, Coeleterates, Reptiles

(1) 3
(2) 4
(3) 5
(4) 6

39) **Assertion :** We can not measure residual volume by spiro meter.
Reason : Volume of air that remain in lungs is consider as RV.

(1) Both assertion and reason are true and the reason is not a correct explanation of the assertion.

(2) Both assertion and reason are true and the reason is a correct explanation of the assertion

(3) Assertion is true but the reason is false

(4) Both assertion and reason are false

40) FRC can be represent by which of the following formula ?

(1) RV + IRV

(2) ERV + RV

(3) TV + ERV

(4) RV + TV + IRV

41) Which of the following capacity is represent by given formula -
Vital capacity + Residual volumue

(1) Inspiratory capacity

(2) Expiratory capacity

(3) Vital capacity

(4) Total lungs capacity

42) pCo_2 is high & Po_2 low is found in which level :-

(1) Alveoli

(2) Blood

(3) Tissue

(4) Heart

43) Which of the following is not a property of cancerous cells ?

(1) Excessive mitosis

(2) Abnormal mitosis

(3) Uncontrolled mitosis

(4) Contact inhibition

44) Among sexually transmitted infection, which of the following can be transmitted by sharing of injection needles and blood transfusion ?

(1) Genital herpes

(2) Gonorrhoea

(3) Trichomoniasis

(4) Hepatitis-B

45) Which of the following statement is **false** ?

(1) Basophils are granular cells

(2) inflammation is a local response of living mammalian tissue to injury

(3) AMI takes part in only innate immunity.

(4) N-K cells kill the target cell by creating perforin lined pores in their cell membrane

46) Which of the following organism is used in production of hepatitis-B vaccine ?

(1) *E.coli*
(2) Sheep
(3) Transgenic yeast
(4) Goat

47) Which of the following statement is correct?

(1) T - lymphocytes are produced in the thymus
(2) B-lymphocytes are produced in thymus.
(3) B- cells produce plasma cell and memory cell.
(4) T-cells do not produce cytokines.

48) If a person shows production of interferons in his body, chances are that he is suffering from :-

(1) Diphtheria
(2) Tetanus
(3) Hepatitis-B
(4) Typhoid



49)
Above diagram is of :-

(1) Datura
(2) Atropa belladona
(3) Opium
(4) Cannabis

50) Which of the following gene is absent in a normal cell ?

(1) Proto-oncogene
(2) Gene related to programme cell death
(3) Tumor suppressor gene
(4) Oncogene

51) Which of the following is not a side effect of anabolic steroids in male ?
(A) Increased sperm count

(B) Enlargement of prostate gland
(C) Increase in size of testicles
(D) Breast enlargement
(E) Increased aggressiveness

(1) Only A
(2) A and C
(3) A, C and D
(4) B and C

52) Choose the correctly matched pair :

(A)	Smoking	Coronary heart disease, Gastric ulcer, Bronchitis
(B)	Anabolic steroid side-effect in female	Reduction in size of clitoris, Premature baldness
(C)	Intravenous drug abuse	High risk of AIDS and Hepatitis-B
(D)	Alcohol abuse	Liver cirrhosis and damage to nervous system

(1) B, C and D
(2) A, C and D
(3) A, B and C
(4) A and D

53) Match these pathogens with their diseases :-

(A)	*Haemophilus influenzae*	(i)	Malignant malaria
(B)	*Entamoeba histolytica*	(ii)	Elephantiasis
(C)	*Plasmodium falciparum*	(iii)	Pneumonia
(D)	*Wuchereria malayi*	(iv)	Typhoid
(E)	*Salmonella typhi*	(v)	Amoebiasis

	A	**B**	**C**	**D**	**E**
(1)	v	i	iii	ii	iv
(2)	iii	i	iv	v	ii
(3)	iii	iv	i	v	ii
(4)	iii	v	i	ii	iv

(1) 1
(2) 2
(3) 3
(4) 4

54) Match the disease in column-I with the appropriate items (pathogen) in column-II:

Column-I		Column-II	
(A)	Syphilis	(i)	*Herpes simplex* virus
(B)	Gonorrhoea	(ii)	*Chlamydia trachomatis*
(C)	Chlamydiasis	(iii)	*Treponema pallidum*
(D)	Genital herpes	(iv)	*Neisseria gonorrhoeae*

Options :

(1) A→ii , B→iii , C→iv , D→i

(2) A→iii , B→iv , C→ii , D→i

(3) A→iii , B→ii , C→iv , D→i

(4) A→ii , B→i , C→iv , D→iii

55) Match column I with column II

Column-I		Column-II	
(i)	BCG	(a)	Passive immunity
(ii)	Cholera	(b)	Toxoid
(iii)	Tetanus	(c)	Inactivated vaccine
(iv)	Anti serum	(d)	Attenuated vaccine

(1) i-d, ii-c, iii-b, iv-a

(2) i-c, ii-d, iii-a, iv-b

(3) i-b, ii-a, iii-d, iv-c

(4) i-a, ii-c, iii-b, iv-d

56) **Assertion :-** Patients are given substances called biological response modifiers such as α-interferon which activates their immune system and helps in destroying the tumor.
Reason :- Tumor cells have been shown to avoid detection and destruction by immune system.

(1) Both Assertion and Reason are True and the Reason is a correct explanation of the Assertion.

(2) Both Assertion and Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion and Reason are False.

57) How many of following disease is associated with cigarette smoking ?
Lung cancer, Throat cancer, Bronchitis, Emphysema, Gastric ulcer, amnesia, Psychosis

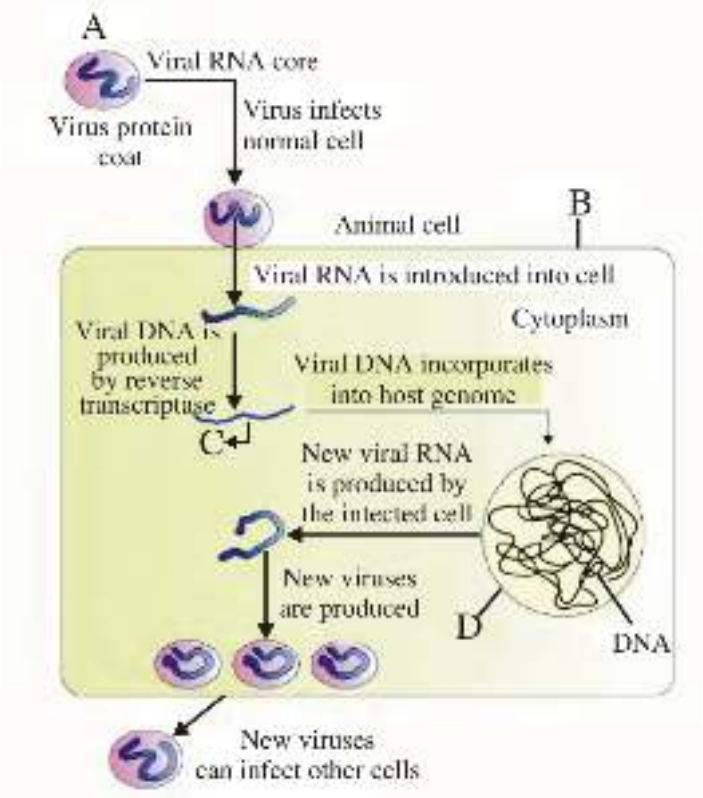
(1) Four

(2) Five

(3) Six

(4) Seven

58) Select the correct answer about the parts labelled as A, B, C and D from the figure given below.



(1) A - Retrovirus, B - Viral DNA,
C - Plasma membrane, D - Nucleus

(2) A - Nucleus, B - Viral RNA,
C - Retrovirus, D - Plasma membrane

(3) A - Retrovirus, B - Plasma membrane,
C - Viral DNA, D - Nucleus

(4) A - Plasma membrane, B - Viral DNA,
C - Nucleus, D - Retrovirus

59) **Assertion (A) :-** Grafts from just any source - an animal, another primate, or any human beings can not be made since the grafts would be rejected sooner or later.
Reason (R) :- The body is able to differentiate 'Self' and 'nonself'.
Choose the correct answer from the given options :-

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(2) (A) is correct but (R) is not correct

(3) Both (A) and (R) are incorrect

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

60) Various disease can spread through inhaling the droplets released by infected person. From the option given below choose the disease that are aquired by this method.
(a) Diptheria
(b) Pneumonia
(c) Dengue
(d) Malaria
(e) Dysentery

(f) Chikunguniya

(1) d,e,f
(2) a,b,c
(3) a and b
(4) c and f

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	3	1	2	2	3	4	4	1	1	2	3	3	2	1	4	2	3	4	4	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	2	4	2	2	4	2	2	3	3	2	4	3	2	3	2	3	2	2	1
Q.	41	42	43	44	45															
A.	4	4	1	2	1															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	2	3	4	1	4	2	2	2	1	2	2	3	2	3	2	2	4	2	3	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	2	2	3	4	1	4	4	4	4	1	2	2	1	3	3	2	4	2	1	2
Q.	86	87	88	89	90															
A.	3	3	2	2	4															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	4	1	3	1	4	4	3	4	4	3	2	3	3	2	3	1	2	4	2	4
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	3	1	4	3	4	3	4	3	3	2	2	4	1	4	2	3	4	1	2	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	3	4	4	3	3	3	3	1	4	2	2	4	2	1	1	2	3	4	3

SOLUTIONS

PHYSICS

1) The resistors of 4 Ω and 8 Ω are connected in parallel.
 Potential drop across 4 Ω resistor =
Potential drop across 8 Ω resistor
Let current in 4 Ω resistor be i
 4i = 8 × 2
or i = 4 A
 current through 2Ω resistor
= 4 + 2 = 6 A

2) $R = \frac{\rho \ell}{A}$

$R' = \frac{\rho' \ell'}{A'}$

$= \frac{\rho (1 + \alpha_\rho \Delta T) . \ell (1 + \alpha_\ell \Delta T)}{A (1 + 2\alpha_\ell \Delta T)}$

$= R \frac{(1 + \alpha_\rho \Delta T) (1 + \alpha_\ell \Delta T)}{(1 + 2\alpha_\ell \Delta T)}$

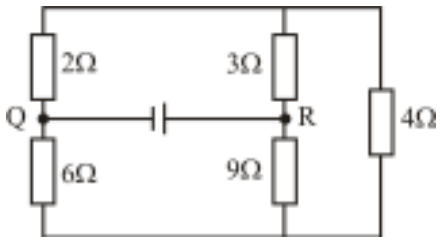
Using, binomial approx.

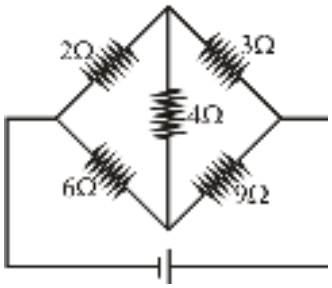
$R' = R (1 + (\alpha_\rho - \alpha_\square)\Delta T)$

Where, $\alpha_\rho = \frac{1}{\rho} \left(\frac{\Delta \rho}{\Delta T} \right)$

$\alpha_\ell = \frac{1}{L} \left(\frac{\Delta L}{\Delta T} \right)$

3)



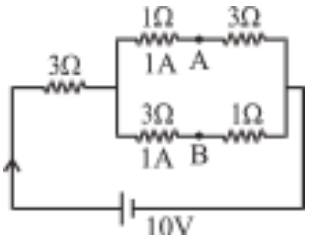


⇒

⇒ wheatstone bridge

⇒ $R_{eq} = \frac{5 \times 15}{5 + 15} = \frac{15}{4}\ \Omega$

4)



$V_A - V_B = -1 \times 1 + 3 \times 1 = 2$ volt

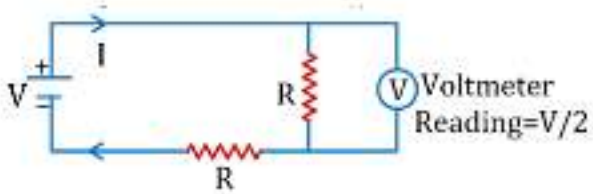
5) $I = I_a\left(\frac{G+S}{S}\right)$

$100 = {}^{1}\left(\frac{99+S}{S}\right)$

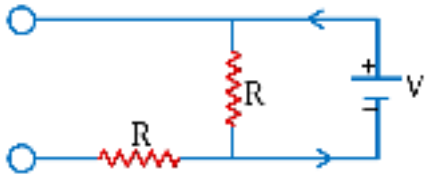
6)

(A) Case (1)

Battery is connected at terminal (I)

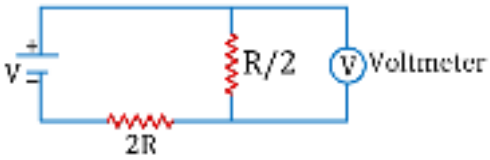


Case:-(2) Battery is connected at terminal (II)



Voltmeter Reading = (V)

All current flow through the resistance (R)

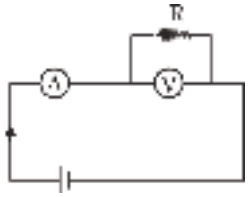


(B)

Reading $\neq$ V/2

(C) Reverse case of (A)

(D) Same as (A)



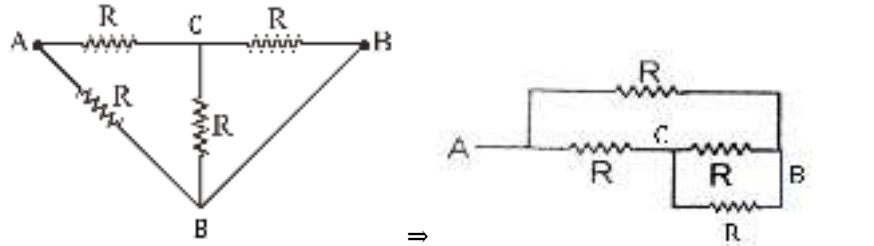
7) when a resistance is joined in parallel with voltmeter, effective resistance of circuit will decrease and current through ammeter increases so, A increases. The sum of potential at A & V is constant so, V decreases.

8) $P = V^2/R$. If P is more, R is less. $R = \rho\ell/a$. For less R, “a” is more . So, the 100W bulb has

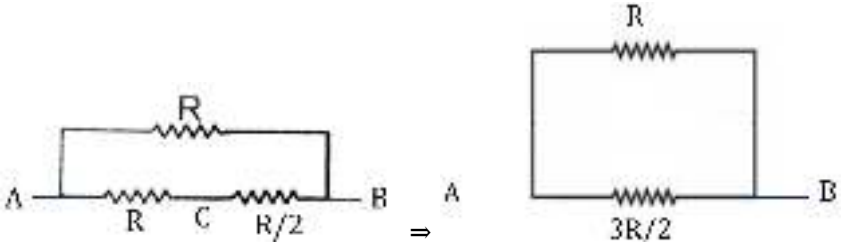
thicker element.

9)

We can redraw circuit as :



⇒



⇒ ⇒

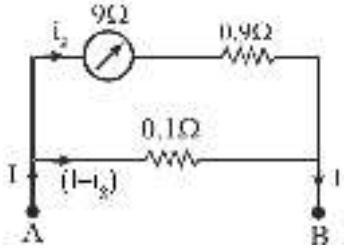
$$\frac{1}{R_{eq}} = \frac{1}{R} + \frac{1}{(3R/2)}$$

$$R_{eq} = \frac{3R}{5}$$

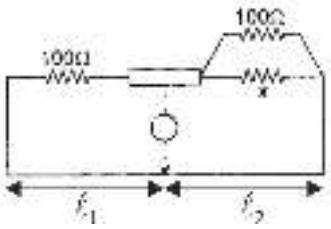
10) Number for yellow is 4, Number of violet is 7 Brown colour gives multiplier 10^1,
Gold gives a tolerance of ± 5%
So resistance of resistor is $47 \times 10^1\ \Omega \pm 5\%$
or $470 \pm 5\%\ \Omega$.

11) Increased temperature leads to more frequent collisions which reduce electron mobility and thus conductivity
number of free electron in a conductior are independent of temperature.

12) $i_g = 10\ mA = 0.01A$
$V_A = V_B = (I - i_g)0.1 = ig \times 9.9$
or $I \times 0.1 = 10i_g$



$$I = \frac{10 \times 0.01}{0.1} = 1\ A$$

13) 

 wheat stone bridge is in balanced condition

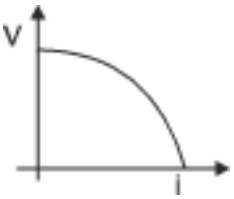
so $\frac{100}{\ell_1} = \frac{100\ x}{100 + x}$

$\frac{\ell_1}{\ell_2}$

 $\frac{\ell_1}{\ell_2} = 2 \Rightarrow x = 100\ W$

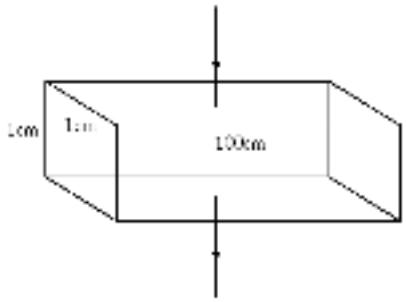
14) Red – 2 (1st digit)
Yellow – 4 (2nd digit)
Voilet – 7 (multiplier) $\Rightarrow 10^7$
Silver – 10% tolerance

15) Given $r \propto i$
$\Rightarrow r = ki$
$V = E - ir = E - i(ki)$



$V = -i^2 k + E$

16) $V \propto I^2$
$P \propto VI \propto I^3$



17)

$$R = \rho\frac{\ell}{A} = \frac{3 \times 10^{-7} \times (1 \times 10^{-2})}{100 \times 1 \times 10^{-4}}$$

$= 3 \times 10^{-7}\ \Omega$

18) **Explain Question:** A trigonometric expression is given and it is asked to find approx. vale using small angle approximation
Concept: Small angle approximation
Solution:

$\frac{\sin 1^\circ}{\cos 2^\circ}$

For small angle

$\sin\theta \approx \theta$

$\cos\theta \approx 1$

$\frac{\sin 1^\circ}{\cos 2^\circ} = \frac{1 \times \frac{\pi}{180}}{1}$

$= \frac{\pi}{180}$

Final answer:

Approximate values is

$= \frac{\pi}{180}$

19) $(x + y)^2 = 4$

$x + y = \pm 2$

$y = \pm 2 - x$

$\frac{dy}{dx} = 0 - 1$

$= -1$

20) $y_{min} = \frac{-D}{4A} = \frac{-1}{4}$

$(x - 7)(x - 8) = x^2 + 15x + 56 = 0$

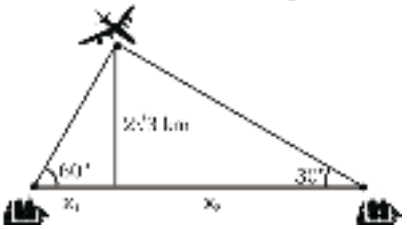
$x = 7, 8$

$x = \frac{-b}{2a} = \frac{+15}{2} = 7.5$

21) $\frac{2\sqrt{3}}{x_1} = \tan 60^\circ$

& $\frac{2\sqrt{3}}{x_2} = \tan 30^\circ \Rightarrow x_1 = 2\text{ km},\ x_2 = 6\text{ km}$

Distance between ships = $x_1 + x_2 = 2 + 6 = 8$ km



22) $\int_0^{\pi/2} \cos 3t\, dt = \left[\frac{\sin 3t}{3}\right]_0^{\pi/2}$

$= \frac{1}{3}[-1 - 0] = \frac{-1}{3}$

23)

Possible lengths would be $|\vec{A} + \vec{B}|$ or $|\vec{A} - \vec{B}|$

24)

(P) $\vec{a} \cdot \vec{b} = ab\cos\theta = 0$

$\therefore \theta = \frac{\pi}{2}$

(Q) $\hat{a} \cdot \vec{b} = b\cos\theta = 4$

$\Rightarrow 4\cos\theta = 4$

$\therefore \theta = 0$

(R) $\vec{a} \cdot \vec{b} = -12 = 3 \times 4\cos\theta$

$\therefore \theta = \pi$

(S) $\left|\vec{a} + \vec{b}\right| = 5 = \sqrt{3^2 + 4^2}$

$\therefore \theta = \frac{\pi}{2}$

25) $\vec{A} = 40 \uparrow$

$\vec{B} = -25\cos 37^\circ \uparrow + 25\sin 37^\circ \uparrow$

$\vec{B} = -20 \uparrow + 15 \uparrow$

$\vec{A} + \vec{B} = 20 \uparrow + 15 \uparrow$

$\tan\theta_x = \frac{15}{20} = \frac{3}{4}$

θ_x = 37° north of east

26) V = 10 m/sec

$\hat{V} = \frac{3\hat{i} + 4\hat{j}}{5}$

$\vec{V} = v\hat{v}$

$\vec{V} = (10)\left(\frac{3\hat{i} + 4\hat{j}}{5}\right) = 6\hat{i} + 8\hat{j}$

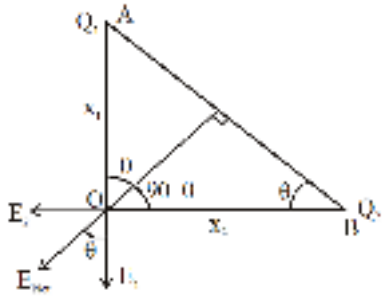
27) $(\vec{A} \times \vec{B}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & 3 & 2 \\ 6 & 8 & 4 \end{vmatrix} = 4\hat{i} + 4\hat{j} + 14\hat{k}$

$(\vec{A} \times \vec{B}) \cdot \vec{C} = (4\hat{i} + 4\hat{j} + 14\hat{k}) \cdot (3\hat{i} + \hat{j} - \hat{k})$

= 12 + 4 – 14 = 2

28)

Due to symmetry of charge distribution in III the net electric field is zero.



29)

E_2 = electric field due to Q_2

$$= \frac{kQ_2}{x_2^2}$$

$$E_1 = \frac{kQ_1}{x_1^2}$$

From diagram

$$\tan\theta = \frac{E_2}{E_1} = \frac{x_1}{x_2}$$

$$\frac{kQ_2}{x_2^2 \times \frac{kQ_1}{x_1^2}} = \frac{x_1}{x_2}$$

$$\frac{Q_2 x_1^2}{Q_1 x_2^2} = \frac{x_1}{x_2}$$

$$\frac{Q_2}{Q_1} = \frac{x_2}{x_1}$$

$$\frac{Q_1}{Q_2} = \frac{x_1}{x_2}$$

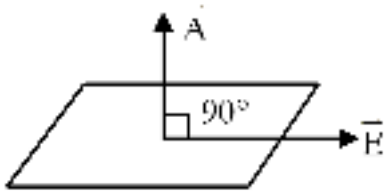
30) (i) is incorrect because '1' and '2' are replelled by '3' so '1' and '2' have charge of same nature. Hence they should repel each other. For this reason, (ii) is correct.
(iii) is correct if charge on '2' and '3' is of same nature and on '1' opposite of them.
(iv) All attraction is not possible.

31) Place another identical hemisphere on given one, so that flat faces of two coincide The total flux linked with cross- section of bottom half (lower hemisphere) is $\frac{Q}{2\varepsilon_0}$ as charge enclosed by given hemisphere is zero. So flux linked with flat face of hemisphere is $\frac{-Q}{2\varepsilon_0}$

32) $\oint \vec{E} \cdot d\vec{s} = \frac{q_{in}}{\varepsilon_0}$

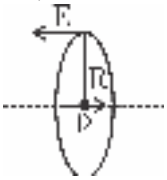
E = EF due to all charges

33) Since $\phi = \vec{E}.\vec{A} = EA\cos\theta$



$\theta = 90°$
$\therefore \phi = 0$

34) Electric field at each point on the surface of ring due to dipole is $E = \frac{kp}{R^3}$



in direction opposite to the dipole moment. (figure below)

Hence netforce on ring is $F = QE = \frac{kpQ}{R^3}$

35) Electric field inside conductor is always zero in electrostatic condition & if electric field is zero then all point will have same potential

as $\Delta V = -\int \vec{E} \cdot d\vec{r}$

So if $\vec{E} = 0$ then $\int \vec{E} \cdot d\vec{r} = 0$ then $\Delta V = 0$

So there will be no potential difference, all point will have same potential.

36) For (A) : Torque $\vec{\tau} = \vec{p} \times \vec{E}$

For (B), (C) & (D) : Potential energy $U = -\vec{p} \cdot \vec{E}$

37) Electric field inside the conductor due to the charges placed in the cavity will be cancelled only by charges induced on the inside surface (shielding effect)

38) $E = -\frac{dv}{dr}\hat{r} = \frac{343}{7^3}\left(3\hat{i} + 2\hat{j} + 6\hat{k}\right)$

39) P.E. of an electron, U= –eV

V = potential at that point

V is negative maximum at point B, hence U is maximum for this point.

40)

Let charge at (–1) be q_1 & at (+1) be q_2.

$\therefore \frac{Kq_1}{0.5} + \frac{Kq_2}{1.5} = \frac{Kq_1}{1} + \frac{Kq_2}{1}$ [V = 2.7 volt]

$\Rightarrow 2q_1 + \frac{2}{3}q_2 = q_1 + q_2$

$\Rightarrow q_1 = \frac{q_2}{3} \Rightarrow \frac{q_1}{q_2} = \frac{1}{3}$

Both the charges are positive.

41)

$\Delta U = qV = 80 \times 4 \times 10^6$ J

42) $V = -\frac{kQ^2}{a} \times 5 + \frac{kQ^2}{a} \times 7$
$-\frac{kQ^2}{\sqrt{2}a} \times 7 + \frac{kQ^2}{\sqrt{2}a} \times 5 + \frac{2kQq}{\frac{\sqrt{3}a}{2}}$
$= \frac{2kQ^2}{a} - \frac{2kQ^2}{\sqrt{2}a} + \frac{4kQq}{\sqrt{3}a}$

43) In electrostatic condition electric field is perpendicular to equipotential surface and surface of conductor is equipotential.

44) Minimum charge is charge of electrons we can't have small charge then this

45) $U_2 = U_1 + qV \Rightarrow V = \frac{U_2 - U_1}{q}$

CHEMISTRY

46)

In x $\Rightarrow O_2^{-2}$ is present
In y $\Rightarrow O_2^{-1}$ is present
 BE $O_2^- > O_2^{2-}$
y > x

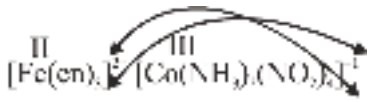
47)

$S_1 \rightarrow [Co(ox)_3]^{3-} \rightarrow$ SFL
$S_2 \rightarrow [Co(NH_3)_3Cl_3]$
$S_3 \rightarrow$ Possible isomers $\Rightarrow$ 6
$NO_2 \rightarrow$ Ambidentate ligend
$S_4 \rightarrow$ Fe oxidation state $\rightarrow +1$

48)

A $\rightarrow [Co(H_2O)_6]Cl_3$
B $\rightarrow [Co(H_2O)_5Cl].Cl_2.H_2O$

C $\rightarrow [Co(H_2O)_4Cl_2]Cl.\ 2H_2O$
Shows G.I.



49) $[Fe(en)_3]^{2+}$ $[Co(NH_3)_2(NO_2)_4]^-$

50)

Asking for:

Determine the relationship between the assertion and reason regarding metal carbonyl bonding.

Concept:

A. **Metal Carbonyls:** Coordination complexes where carbon monoxide (CO) acts as a ligand.
B. **Synergic Effect:** A cooperative strengthening of the metal-ligand bond due to both σ-donation and π-backbonding.
C. **σ and π Characters:** The nature of the metal-carbon bond, involving both sigma and pi interactions.

Formula:

A. General formula for metal carbonyls: M(CO)x (where M is a metal and x is the number of CO ligands).

Solution/Explanation:

A. **Assertion (A):** The assertion is true. In metal carbonyls, a synergic effect occurs. The CO ligand donates electron density to the metal (σ-donation), and the metal, in turn, back-donates electron density to the CO ligand's antibonding π* orbitals (π-backbonding). This strengthens the metal-carbon bond.
B. **Reason (R):** The reason is also true. The metal-carbon bond in metal carbonyls does indeed possess both σ and π characters. The σ bond is formed by the donation of electrons from the lone pair present on carbon to the metal. The π bond is formed by the back donation of electrons from filled d-orbitals of metal to the empty antibonding π* orbitals of CO.
C. **Relationship:** The reason (R) directly explains the assertion (A). The presence of both σ and π character in the metal-carbon bond is the basis of the synergic effect, which strengthens the bond.

Therefore, both the assertion and reason are true, and the reason is the correct explanation of the assertion.

Answer : Option : 4

51)

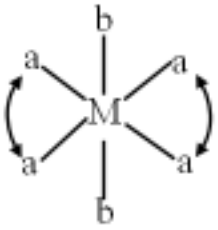
NCERT Pg. No. 245

52)

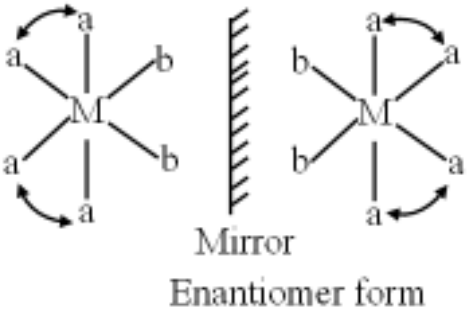
$[Ni(dmg)_2]$ having 4 rings

53)

$[Cu(H_2O)_6]^{+2}$
Hyb ⇒ sp^3d^2
CN → 6
→ 1 unpaired e^- are present.



54)
P.O.S present so optically inactive



Enantiomer form

55)

$K_2[Cr(CN)_2 \quad O_2 \quad (O_2) \quad NH_3]$
↓ ↓ ↓ ↓
di Cynido dioxygen dioxo Ammine
(–1) (O_2 = 0) (O^{-2})

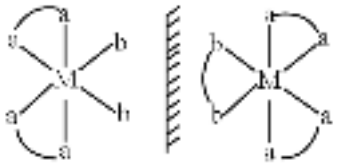
56)

$\Delta_{sp} > \Delta_o > \Delta_t$

57)

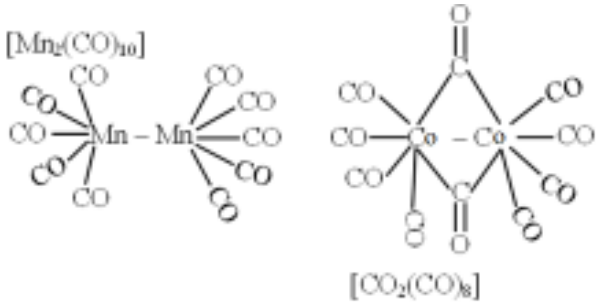
$[Mn(CN)_6]^{-3}$
⇓
d^2sp^3 (Hyb)
Mn^{+3} → [Ar] $3d^4$
$[Fe(CN)_6]^{-3}$
⇓
d^2sp^3 (Hyb)
Fe^{+3} ⇒ [Ar]$3d^5$

58)



Cis-form (Enantiomers)

59)



60)

Cyanide process → $[Ag(CN)_2]$–
Mond's process → $Ni(CO)_4$
Photographic fixing process → $[Ag\ (S_2O_3)_2]^{-3}$

61)

Inert gases generally chemically unreactive Xe show variable oxidation state.

62) Except Nitrogen, In group 16 all elements show Allotropy.

63)

N does not show pentavalent covalency.
NF_5 → does not exist

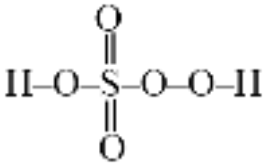
64)

$$2H_2SO_4 \xrightarrow{-H_2O} H_2S_2O_7$$

Pyrosulphuric acid

65)

Caro's acid



66)

Al_2Cl_6 overlapping ⇒ $sp^3 – sp^3 – sp^3$

67) $PCl_5 + H_2O \rightarrow POCl_3 + 2HCl$
$POCl_3 + 3H_2O \rightarrow H_3PO_4 + 3HCl$

_____

net rxn. $PCl_5 + 4H_2O \rightarrow H_3PO_4 + 5HCl$

_____

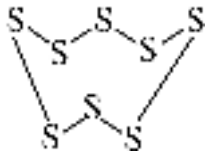
A = PCl_5, B = $POCl_3$, C = H_3PO_4

68)

(A) $(BeH_2)_2$ → 3 centre $–2e^-$
(B) $Al_2(CH_3)_6$ → 3 centre $–2e^-$
(C) $(BeCl_2)_2$ → 3 centre $–4e^-$
(D) $(AlCl_3)_2$ → 3 centre $–4e^-$

69)

Both are allotropes of sulphur



Crown shape (S_8) Monoclinic sulphur are soluble in CS_2 but insoluble in water.

70)

SRP value
$F_2(E^0 = +2.87\ V) > Cl_2\ (E^0 = +1.36\ V)$
I_2O_5 → Oxidising agent

71) NCERT Page No.# 310
Atomic Radius : B < Ga < Al < In < Tl
Ionisation Energy : B > Tl > Ga > Al > In
Electronegativity : B > Tl > In > Ga > Al

72) NCERT Pg # 200

73)

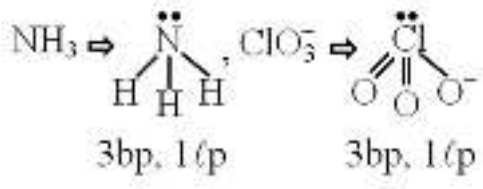
BP ⇒ $O_2 < S_8$
Reducing nature ⇒ $H_2O < H_2Se$

74)

(1) H_2O have high MP & BP due to H-bonding.
(2) In case of hydride from top to bottom acidic strength increases.

75)

See module Pg. # 33

76)


both are trigonal pyramidal

77)

Bond angle in Cl_2O is more than 109°28' due to large size of Cl, πp-πp repulsion is more

78)

Ionic character – $CdCl_2 < SnCl_2 < CaCl_2$
Polarisation power of $Cd^{+2} > Sn^{+2} > Ca^{+2}$
(outer most e^-s) $(18e^-)$ $(18+2)e^-$ $(8e^-)$

79)

NO	→	NO^+
$15e^-$		$14e^-$
B.O. = 2.5		B.O. = 3
Unpaired		Unpaired
e^- = 1		e^- = 0
(para)		(dia)

80)

H_3C –$CH(OH)_2$ is unstable due to steric repulsion.

81)

(A) PH_5, NCl_5, $BiCl_5$ not exist
(B) I_3^+, sp^3, 2bp, 2lp bent
(C) Dipole moment of XeF_4 is zero due to symmetrical molecule.

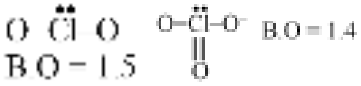
82)  bent

83)

(SiC) Covalent solid has very high M.P.

84)

$ClO^- < ClO_2^- < ClO_3^-$

Cl – O^- B.O = 1 

85) XeF_3^+, XeF_5^+
sp^3d sp^3d^2
3bp, 2lp 5bp, 1lp
Bent'T' square pyramid

86)

Conceptual (NCERT)

87)

XeF_4, B= bp = 4
(sp^3d^2) E = lp = 2
Square plannar

88) **Explanation** -
The question asks to identify the incorrect sequence of bond dissociation energies among the given sets of bonds, based on bond strength.

Concept - (Bond dissociation energy & bond strength)

A. Bond dissociation energy decreases with increasing bond length and lone pair repulsion. H-H is strongest, followed by C-C, then Si-Si.
B. C-C > N-N > O-O is correct due to lone pair repulsions.
C. H-Cl > H-Br > H-I is correct due to decreasing bond strength.
D. Therefore, Si-Si > C-C > H-H is the incorrect order.

Final Answer - Option (2)

89)

H_2SO_4 has intermolecular H-Bonding (more than HNO_3)

90)

Explain Question : Match molecular species with correct molecular orbital theory-based properties.

Concept : This question is based on MOT

Solution :

(A) $\mathbf{N_2}$ **:** (14 electrons) - $\sigma 1s^2$, σ^*1s^2, $\sigma 2s^2$, σ^*2s^2, $\pi 2px^2 = \pi 2py^2$, $\sigma 2pz^2$ B.O. = 3

$\mathbf{N_2^+}$ **:** (13 electrons) - $\sigma 1s^2$, σ^*1s^2, $\sigma 2s^2$, σ^*2s^2, $\pi 2px^2 = \pi 2py^2$, $\sigma 2pz^1$ B.O. = 2.5

(B) $\mathbf{B_2}$ **:** (10 electrons) - $\sigma 1s^2$, σ^*1s^2, $\sigma 2s^2$, σ^*2s^2, $\pi 2px^1 = \pi 2py^1$ B.O. = 1

$\mathbf{B_2^+}$ **:** (9 electrons) - $\sigma 1s^2$, σ^*1s^2, $\sigma 2s^2$, σ^*2s^2, $\pi 2px^1$ B.O. = 0.5

(C) $\mathbf{O_2}$ (16 electrons) - $\sigma 1s^2$, σ^*1s^2, $\sigma 2s^2$, σ^*2s^2, $\sigma 2pz^2$, $\pi 2px^2 = \pi 2py^2$, $\pi^*2px^1 = \pi^*2py^1$ B.O. = 2

$\mathbf{O_2^{2-}}$ (18 electrons) - $\sigma 1s^2$, σ^*1s^2, $\sigma 2s^2$, σ^*2s^2, $\sigma 2pz^2$, $\pi 2px^2 = \pi 2py^2$, $\pi^*2px^2 = \pi^*2py^2$ B.O. = 1

(D) $\mathbf{H_2}$ (2 electrons) - $\sigma 1s^2$ B.O. = 1

$\mathbf{H_2^+}$ (1 electrons) - $\sigma 1s^1$ B.O. = 0.5

Final Answer : option (4)

BIOLOGY

91)

NCERT XII, Pg. # 124, 125

92)

NCERT XII, Pg. # 111, 118,119

93)

NCERT XII, Pg. # 116

94) NCERT XII, Pg. # 113

95)

Module - Evolution

96)

NCERT-XII, Pg. # 116

97) NCERT XII, Pg. # 124,125

98) NCERT XII, Pg. # 122

99)

NCERT XII, Pg. # 113

100)

NCERT XII, Pg. # 125

101)

NCERT XII, Pg. # 117

102)

NCERT XII, Pg. # 111

103)

NCERT XII, Pg. # 121

104)

NCERT XII, Pg. # 121

105)

NCERT XII, Pg. # 116

106)

NCERT XII, Pg. # 114,115

107) NCERT XII, Pg. # 125

108) NCERT XII, Pg. # 121

109)

NCERT XI Page No. # 283

110)

NCERT Pg No. # 281 (Para-1)

111) NCERT XI Pg # 279, 280

112) NCERT Pg#280

113)

Bio Module # 5, Pg. 104

114) NCERT, Pg. # 280

115) NCERT-XI, Pg. # 279-280

116) **Question Explanation :**
TO IDENTIFY THE CORRECTLY MATCHED STRUCTURES

Concept :
WBC

Solution:
Monocytes has Kidney shaped nucleus

Final Answer :
Option (3)

117) NCERT-XI , Pg# 285

118)

Explain Question: to identify the medical condition in which there is Pain in heart muscles

Concept: disorder of circulatory system

Solution: Angina pectoris is the medical term for chest pain that results from inadequate oxygen supply to the heart muscle.

Final answer: option (3)

119) NCERT XI Page No. # 284

120) NCERT Pg.# 282, 18.3

121)

NCERT XI, Pg # 187

122) NCERT Pg. # 275

123) NCERT (XI) Pg. # 272

124) NCERT (XI) Pg. # 269

125) NCERT (XI) Pg. # 273

126) NCERT (XI) Pg. # 273

127)

NCERT Pg.# 184

128) NCERT Pg. # 268, Para 2

129) NCERT-XI_Pg. # 187

130) NCERT Pg. # 187

131) NCERT-XI_ Pg. # 187

132) NCERT-XI_Pg. # 187

133) NCERT (XII) Pg. # 157, Para-8.4

134)

Explain Question : Identify the sexually transmitted infection that can also be transmitted via injection needles and blood transfusion.

Concept : This Question is based on STDs

Solution : The correct answer is option 4.

A. Hepatitis-B is a viral infection transmitted not only sexually but also through contaminated needles and infected blood. The other options are primarily limited to sexual transmission.

Final answer : 4

135) NCERT XII Pg. # 151

136) NCERT-XII, Page# 152

137)

The correct answer is Option 3: B-cells produce plasma cells and memory cells.
Here's why:
• B cells are a type of white blood cell that play a key role in the adaptive immune response. When activated by an antigen, B cells differentiate into two primary types:
o Plasma cells: These cells produce antibodies, which are proteins that bind to specific antigens and neutralize them.
o Memory cells: These cells "remember" the antigen and can quickly respond if the body encounters it again in the future, providing long-lasting immunity.
Let's look at the other options:
• **Option 1:** T-lymphocytes are produced in the thymus. This is partially correct. T lymphocytes are indeed produced in the bone marrow, but they mature and differentiate in the thymus.
• **Option 2:** B-lymphocytes are produced in the thymus. This is incorrect. B lymphocytes are produced in the bone marrow.
• **Option 4:** T-cells do not produce cytokines. This is incorrect. T cells can produce cytokines, which are signaling molecules that help regulate the immune response.
Therefore, the statement that is most accurate is Option 3: B-cells produce plasma cells and memory cells.

138) NCERT Page-151

139) NCERT (E) Pg. # 159

140)

NCERT Pg. # 157

141) **Explanation:**
Anabolic steroids are synthetic hormones that mimic testosterone, promoting muscle growth but causing side effects. In males, they disrupt natural hormone balance, affecting reproductive and physical health.
Correct Answer:
Option 2: A and C (Increased sperm count and increase in size of testicles are NOT side effects).
Explanation:
Steroids **decrease sperm count** by suppressing natural testosterone production, leading to **testicular shrinkage (atrophy)** instead of enlargement. Other effects include **prostate enlargement, gynecomastia (breast growth),** and **aggression** due to hormonal imbalance.

142)

NCERT-XII, Pg. # 146, 144

143)

NCERT-XI Pg. # 148,149 (E)
NCERT-XI Pg. # 159,160 (H)

144) NCERT (XIIth) Pg. # 63

145) NECER-XII, Pg # 152

146) NCERT Pg. # 142

147) NCERT (XIIth) Pg. # 160 Para 8.5

148) NCERT-XII, Pg. # 155, Para-8.3, 8..6

149)

The correct answer is Option 4.
Analysis:

A. Both Assertion (A) and Reason (R) are correct.
B. **Assertion (A):** This is true. The immune system of the recipient can recognize the transplanted tissue as "foreign" and initiate an immune response, leading to rejection. This is why tissue matching is crucial in organ transplantation.
C. **Reason (R):** This is also true. The immune system has the ability to distinguish between "self" (the body's own cells and tissues) and "nonself" (foreign substances like bacteria, viruses, and transplanted tissues). This self-recognition is a fundamental aspect of the immune response.
D. Reason (R) is the correct explanation of Assertion (A).
E. The rejection of grafts is directly related to the body's ability to differentiate between "self" and "nonself." If the transplanted tissue is recognized as "nonself," the immune system will mount an attack against it.

Therefore, both the assertion and reason are correct, and the reason provides the correct explanation for the assertion.

150) NCERT Pg # 130,131,132